

Trabalho 1 — Análise de DNS: Censura, Desempenho e Privacidade

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Parte 1 — Scanner DNS sobre UDP

1.1 Análise de Bloqueio

A tabela abaixo compara as respostas dos servidores UDP para os domínios testados. Legenda: OK = NOERROR, NXDM = NXDOMAIN, DIV = IP divergente do consenso, NULO = endereço nulo (0.0.0.0).

Domínios — D1: www.example.com | D2: www.pucrs.br | D3: internetbadguys.com | D4: reddit.com | D5: tinder.com | D6: polymarket.com | D7: www.google.com | D8: www2.thepiratebay3.co | D9: isitblocked.org

Servidor	D1	D2	D3	D4	D5	D6	D7	D8	D9
Google (8.8.8.8)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Google 2 (8.8.4.4)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Cloudflare (1.1.1.1)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Cloudflare 2 (1.0.0.1)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Quad9 s/ filtro (9.9.9.10)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Verisign (64.6.64.6)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Quad9 Seg. (9.9.9.9)	OK	OK	OK	OK	OK	OK	OK	OK	NXDM
OpenDNS (208.67.222.222)	OK	OK	DIV	OK	OK	OK	OK	OK	OK
CleanBrow. Seg. (185.228.168.9)	OK	OK	NXDM	OK	OK	OK	OK	OK	OK
AdGuard DNS (94.140.14.14)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Cloudflare Family (1.1.1.3)	OK	OK	OK	OK	OK	OK	DIV	NULO	OK
OpenDNS Family (208.67.222.123)	OK	OK	DIV	OK	OK	OK	OK	OK	OK
CleanBrow. Family (185.228.168.168)	OK	OK	NXDM	NXDM	NXDM	OK	DIV	OK	OK
AdGuard Family (94.140.14.15)	OK	OK	OK	OK	DIV	OK	DIV	OK	OK
Control D (76.76.2.0)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Comodo Secure (8.26.56.26)	OK	OK	OK	OK	OK	OK	OK	OK	OK
OpenDNS 2 (208.67.220.220)	OK	OK	DIV	OK	OK	OK	OK	OK	OK
Quad9 ECS (9.9.9.11)	OK	OK	OK	OK	OK	OK	OK	OK	NXDM
AdGuard Unfilt. (94.140.14.140)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Yandex (77.88.8.8)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Sistema (gateway)	OK	OK	DIV	OK	OK	OK	DIV	OK	OK

Técnicas identificadas:

- NXDOMAIN — CleanBrowsing Security bloqueou internetbadguys.com. CleanBrowsing Family bloqueou internetbadguys.com, reddit.com e tinder.com. Quad9 (9.9.9.9 e 9.9.9.11) bloqueou isitblocked.org por NXDOMAIN. Nenhum servidor bloqueou www2.thepiratebay3.co por NXDOMAIN.
- IP Divergente / Redirecionamento — OpenDNS (208.67.222.222, .123 e .220) retornou IP de página de aviso para internetbadguys.com. Cloudflare Family, CleanBrowsing Family e AdGuard Family redirecionaram www.google.com para o endpoint de SafeSearch forçado (216.239.38.120). AdGuard Family redirecionou tinder.com para 94.140.14.35. O gateway doméstico também redirecionou www.google.com e internetbadguys.com.
- Endereço nulo — Cloudflare Family (1.1.1.3) retornou 0.0.0.0 para www2.thepiratebay3.co. Todos os demais servidores resolveram o domínio normalmente.
- polymarket.com — nenhum servidor bloqueou. O bloqueio judicial da Anatel não está implementado nos servidores testados.

Há consenso dentro das categorias: servidores de filtragem familiar bloquearam os mesmos domínios de conteúdo adulto. Servidores sem filtragem não bloquearam nada além dos domínios com filtragem de segurança específica. O bloqueio DNS é trivialmente contornável trocando o servidor configurado no sistema.

Tabela de bloqueio — DNS over TLS:

Servidor	D1	D2	D3	D4	D5	D6	D7	D8	D9
Google DoT (dns.google)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Cloudflare DoT (one.one.one.one)	OK	OK	OK	OK	OK	OK	OK	OK	OK
Quad9 DoT (dns.quad9.net)	OK	OK	OK	OK	OK	OK	OK	OK	NXD M

Os servidores DoT apresentaram comportamento equivalente aos seus pares UDP sem filtragem (Google, Cloudflare) e com filtragem de segurança (Quad9 bloqueou isitblocked.org). Nenhum DoT bloqueou www2.thepiratebay3.co.

1.2 Análise de Desempenho

90 consultas por servidor UDP (10 por domínio × 9 domínios). Valores em ms. Ranking por tempo médio:

Servidor	Suces so	Mín (ms)	Méd (ms)	Máx (ms)
Google (8.8.8.8)	90/90	52,7	149,7	409,6
Google 2 (8.8.4.4)	90/90	39,3	163,0	1466,6
Cloudflare (1.1.1.1)	90/90	14,4	44,9	700,7
Cloudflare 2 (1.0.0.1)	90/90	12,5	27,0	167,3
Quad9 s/ filtro (9.9.9.10)	90/90	12,5	28,1	256,0
Verisign (64.6.64.6)	90/90	142,1	205,5	407,4
Quad9 Seg. (9.9.9.9)	90/90	12,2	66,3	104,5
OpenDNS (208.67.222.222)	90/90	35,0	107,6	1127,7
CleanBrow. Seg. (185.228.168.9)	89/90	134,8	170,3	342,5
AdGuard DNS (94.140.14.14)	90/90	137,0	254,7	4194,2
Cloudflare Family (1.1.1.3)	90/90	90,7	109,1	351,2
OpenDNS Family (208.67.222.123)	90/90	43,0	159,0	1228,1

No.	Time	Source	Destination	Protocol
20	1.052947657	guileo.local	dns.google	DNS
21	1.060860163	guileo.local	dns.google	DNS
22	1.061032429	guileo.local	dns.google	DNS
23	1.061169517	guileo.local	dns.google	DNS
24	1.106744478	dns.google	guileo.local	DNS
25	1.106745219	dns.google	guileo.local	DNS
26	1.113607383	dns.google	guileo.local	DNS
27	1.139201327	dns.google	guileo.local	DNS

Figura 2 — Query A para ads.google.com. Além do par principal (linhas 20 e 27), o SO disparou 3 consultas PTR reversas automáticas (linhas 21–23), respondidas com "No such name" (linhas 24–26). IP retornado: 172.217.29.174.

Além do par query/response principal, o sistema operacional disparou automaticamente 3 consultas PTR reversas para os endereços IPv6 recebidos — tráfego não originado pelo código. Todas retornaram "No such name". O IP final retornado para ads.google.com foi 172.217.29.174.

Figura 3 — Tráfego DoT capturado pelo Wireshark

7	0.366860636	192.168.1.14	dns.google
9	0.469804200	dns.google	192.168.1.14
10	0.469804440	dns.google	192.168.1.14
13	0.471703722	192.168.1.14	dns.google
14	0.560189259	192.168.1.14	39.224.186.35.bc.googleusercontent.c...
17	0.561361728	192.168.1.14	dns.google
20	0.674260923	dns.google	192.168.1.14
27	0.776350966	39.224.186.35.bc.googleusercontent...	192.168.1.14

TLSv1.3	583	Client Hello (SNI=dns.google)
TLSv1.3	2866	Server Hello, Change Cipher Spec
TLSv1.3	1523	Application Data
TLSv1.3	146	Change Cipher Spec, Application Data
TLSv1.2	109	Application Data
TLSv1.3	118	Application Data
TLSv1.3	678	Application Data, Application Data
TLSv1.2	106	Application Data

Figura 3 — Consulta DoT ao dns.google (porta 853 / TCP). Conteúdo aparece como TLSv1.3 opaco; domínio e IPs não são visíveis.

Figura 4 — Comparação UDP vs. DoT no Wireshark

Queries google.com: type A, class IN Name: google.com [Name Length: 10] [Label Count: 2] Type: A (1) (Host Address) Class: IN (0x0001)	TLSv1.3 Record Layer: Handshake Protocol: Client Hello Content Type: Handshake (22) Version: TLS 1.0 (0x0301) Length: 512
	Handshake Protocol: Client Hello Handshake Type: Client Hello (1) Length: 508 Version: TLS 1.2 (0x0303) Random: d8a28502ac4486e612016c8d21c1fe1b17117978543e5683613 Session ID Length: 32 Session ID: d9e763b015fe4295449825060e51c5990ead380ea756cf Cipher Suites Length: 62 Cipher Suites (31 suites) Compression Methods Length: 1 Compression Methods (1 method) Extensions Length: 373 Extension: server_name (len=15) name=dns.google Extension: ec_point_formats (len=4) Extension: supported_groups (len=22) Extension: session_ticket (len=0) Extension: encrypt_then_mac (len=0) Extension: extended_master_secret (len=0) Extension: signature_algorithms (len=42) Extension: supported_versions (len=5) TLS 1.3, TLS 1.2 Extension: psk_key_exchange_modes (len=2) Extension: key_share (len=38) x25519 Extension: padding (len=201) [JA4: t13d311100_e8f1e7e78f70_d41ae481755e] [JA4_r [...]: t13d311100_002f,0033,0035,0039,003c,003d,0067,006f,0071,0072,0073,0074,0075,0076,0077,0078,0079,0080,0081,0082,0083,0084,0085,0086,0087,0088,0089,0090,0091,0092,0093,0094,0095,0096,0097,0098,0099,0100,0101,0102,0103,0104,0105,0106,0107,0108,0109,0110,0111,0112,0113,0114,0115,0116,0117,0118,0119,0120,0121,0122,0123,0124,0125,0126,0127,0128,0129,0130,0131,0132,0133,0134,0135,0136,0137,0138,0139,0140,0141,0142,0143,0144,0145,0146,0147,0148,0149,0150,0151,0152,0153,0154,0155,0156,0157,0158,0159,0160,0161,0162,0163,0164,0165,0166,0167,0168,0169,0170,0171,0172,0173,0174,0175,0176,0177,0178,0179,0180,0181,0182,0183,0184,0185,0186,0187,0188,0189,0190,0191,0192,0193,0194,0195,0196,0197,0198,0199,0200,0201,0202,0203,0204,0205,0206,0207,0208,0209,0210,0211,0212,0213,0214,0215,0216,0217,0218,0219,0220,0221,0222,0223,0224,0225,0226,0227,0228,0229,0230,0231,0232,0233,0234,0235,0236,0237,0238,0239,0240,0241,0242,0243,0244,0245,0246,0247,0248,0249,0250,0251,0252,0253,0254,0255,0256,0257,0258,0259,0260,0261,0262,0263,0264,0265,0266,0267,0268,0269,0270,0271,0272,0273,0274,0275,0276,0277,0278,0279,0280,0281,0282,0283,0284,0285,0286,0287,0288,0289,0290,0291,0292,0293,0294,0295,0296,0297,0298,0299,0300,0301,0302,0303,0304,0305,0306,0307,0308,0309,0310,0311,0312,0313,0314,0315,0316,0317,0318,0319,0320,0321,0322,0323,0324,0325,0326,0327,0328,0329,0330,0331,0332,0333,0334,0335,0336,0337,0338,0339,0340,0341,0342,0343,0344,0345,0346,0347,0348,0349,0350,0351,0352,0353,0354,0355,0356,0357,0358,0359,0360,0361,0362,0363,0364,0365,0366,0367,0368,0369,0370,0371,0372,0373,0374,0375,0376,0377,0378,0379,0380,0381,0382,0383,0384,0385,0386,0387,0388,0389,0390,0391,0392,0393,0394,0395,0396,0397,0398,0399,0400,0401,0402,0403,0404,0405,0406,0407,0408,0409,0410,0411,0412,0413,0414,0415,0416,0417,0418,0419,0420,0421,0422,0423,0424,0425,0426,0427,0428,0429,0430,0431,0432,0433,0434,0435,0436,0437,0438,0439,0440,0441,0442,0443,0444,0445,0446,0447,0448,0449,0450,0451,0452,0453,0454,0455,0456,0457,0458,0459,0460,0461,0462,0463,0464,0465,0466,0467,0468,0469,0470,0471,0472,0473,0474,0475,0476,0477,0478,0479,0480,0481,0482,0483,0484,0485,0486,0487,0488,0489,0490,0491,0492,0493,0494,0495,0496,0497,0498,0499,0500,0501,0502,0503,0504,0505,0506,0507,0508,0509,0510,0511,0512,0513,0514,0515,0516,0517,0518,0519,0520,0521,0522,0523,0524,0525,0526,0527,0528,0529,0530,0531,0532,0533,0534,0535,0536,0537,0538,0539,0540,0541,0542,0543,0544,0545,0546,0547,0548,0549,0550,0551,0552,0553,0554,0555,0556,0557,0558,0559,0560,0561,0562,0563,0564,0565,0566,0567,0568,0569,0570,0571,0572,0573,0574,0575,0576,0577,0578,0579,0580,0581,0582,0583,0584,0585,0586,0587,0588,0589,0590,0591,0592,0593,0594,0595,0596,0597,0598,0599,0600,0601,0602,0603,0604,0605,0606,0607,0608,0609,0610,0611,0612,0613,0614,0615,0616,0617,0618,0619,0620,0621,0622,0623,0624,0625,0626,0627,0628,0629,0630,0631,0632,0633,0634,0635,0636,0637,0638,0639,0640,0641,0642,0643,0644,0645,0646,0647,0648,0649,0650,0651,0652,0653,0654,0655,0656,0657,0658,0659,0660,0661,0662,0663,0664,0665,0666,0667,0668,0669,0670,0671,0672,0673,0674,0675,0676,0677,0678,0679,0680,0681,0682,0683,0684,0685,0686,0687,0688,0689,0690,0691,0692,0693,0694,0695,0696,0697,0698,0699,0700,0701,0702,0703,0704,0705,0706,0707,0708,0709,0710,0711,0712,0713,0714,0715,0716,0717,0718,0719,0720,0721,0722,0723,0724,0725,0726,0727,0728,0729,0730,0731,0732,0733,0734,0735,0736,0737,0738,0739,0740,0741,0742,0743,0744,0745,0746,0747,0748,0749,0750,0751,0752,0753,0754,0755,0756,0757,0758,0759,0760,0761,0762,0763,0764,0765,0766,0767,0768,0769,0770,0771,0772,0773,0774,0775,0776,0777,0778,0779,0780,0781,0782,0783,0784,0785,0786,0787,0788,0789,0790,0791,0792,0793,0794,0795,0796,0797,0798,0799,0800,0801,0802,0803,0804,0805,0806,0807,0808,0809,0810,0811,0812,0813,0814,0815,0816,0817,0818,0819,0820,0821,0822,0823,0824,0825,0826,0827,0828,0829,0830,0831,0832,0833,0834,0835,0836,0837,0838,0839,0840,0841,0842,0843,0844,0845,0846,0847,0848,0849,0850,0851,0852,0853,0854,0855,0856,0857,0858,0859,0860,0861,0862,0863,0864,0865,0866,0867,0868,0869,0870,0871,0872,0873,0874,0875,0876,0877,0878,0879,0880,0881,0882,0883,0884,0885,0886,0887,0888,0889,0890,0891,0892,0893,0894,0895,0896,0897,0898,0899,0900,0901,0902,0903,0904,0905,0906,0907,0908,0909,0910,0911,0912,0913,0914,0915,0916,0917,0918,0919,0920,0921,0922,0923,0924,0925,0926,0927,0928,0929,0930,0931,0932,0933,0934,0935,0936,0937,0938,0939,0940,0941,0942,0943,0944,0945,0946,0947,0948,0949,0950,0951,0952,0953,0954,0955,0956,0957,0958,0959,0960,0961,0962,0963,0964,0965,0966,0967,0968,0969,0970,0971,0972,0973,0974,0975,0976,0977,0978,0979,0980,0981,0982,0983,0984,0985,0986,0987,0988,0989,0990,0991,0992,0993,0994,0995,0996,0997,0998,0999,1000,1001,1002,1003,1004,1005,1006,1007,1008,1009,1010,1011,1012,1013,1014,1015,1016,1017,1018,1019,1020,1021,1022,1023,1024,1025,1026,1027,1028,1029,1030,1031,1032,1033,1034,1035,1036,1037,1038,1039,1040,1041,1042,1043,1044,1045,1046,1047,1048,1049,1050,1051,1052,1053,1054,1055,1056,1057,1058,1059,1060,1061,1062,1063,1064,1065,1066,1067,1068,1069,1070,1071,1072,1073,1074,1075,1076,1077,1078,1079,1080,1081,1082,1083,1084,1085,1086,1087,1088,1089,1090,1091,1092,1093,1094,1095,1096,1097,1098,1099,1100,1101,1102,1103,1104,1105,1106,1107,1108,1109,1110,1111,1112,1113,1114,1115,1116,1117,1118,1119,1120,1121,1122,1123,1124,1125,1126,1127,1128,1129,1130,1131,1132,1133,1134,1135,1136,1137,1138,1139,1140,1141,1142,1143,1144,1145,1146,1147,1148,1149,1150,1151,1152,1153,1154,1155,1156,1157,1158,1159,1160,1161,1162,1163,1164,1165,1166,1167,1168,1169,1170,1171,1172,1173,1174,1175,1176,1177,1178,1179,1180,1181,1182,1183,1184,1185,1186,1187,1188,1189,1190,1191,1192,1193,1194,1195,1196,1197,1198,1199,1200,1201,1202,1203,1204,1205,1206,1207,1208,1209,1210,1211,1212,1213,1214,1215,1216,1217,1218,1219,1220,1221,1222,1223,1224,1225,1226,1227,1228,1229,1230,1231,1232,1233,1234,1235,1236,1237,1238,1239,1240,1241,1242,1243,1244,1245,1246,1247,1248,1249,1250,1251,1252,1253,1254,1255,1256,1257,1258,1259,1260,1261,1262,1263,1264,1265,1266,1267,1268,1269,1270,1271,1272,1273,1274,1275,1276,1277,1278,1279,1280,1281,1282,1283,1284,1285,1286,1287,1288,1289,1290,1291,1292,1293,1294,1295,1296,1297,1298,1299,1300,1301,1302,1303,1304,1305,1306,1307,1308,1309,1310,1311,1312,1313,1314,1315,1316,1317,1318,1319,1320,1321,1322,1323,1324,1325,1326,1327,1328,1329,1330,1331,1332,1333,1334,1335,1336,1337,1338,1339,1340,1341,1342,1343,1344,1345,1346,1347,1348,1349,1350,1351,1352,1353,1354,1355,1356,1357,1358,1359,1360,1361,1362,1363,1364,1365,1366,1367,1368,1369,1370,1371,1372,1373,1374,1375,1376,1377,1378,1379,1380,1381,1382,1383,1384,1385,1386,1387,1388,1389,1390,1391,1392,1393,1394,1395,1396,1397,1398,1399,1400,1401,1402,1403,1404,1405,1406,1407,1408,1409,1410,1411,1412,1413,1414,1415,1416,1417,1418,1419,1420,1421,1422,1423,1424,1425,1426,1427,1428,1429,1430,1431,1432,1433,1434,1435,1436,1437,1438,1439,1440,1441,1442,1443,1444,1445,1446,1447,1448,1449,1450,1451,1452,1453,1454,1455,1456,1457,1458,1459,1460,1461,1462,1463,1464,1465,1466,1467,1468,1469,1470,1471,1472,1473,1474,1475,1476,1477,1478,1479,1480,1481,1482,1483,1484,1485,1486,1487,1488,1489,1490,1491,1492,1493,1494,1495,1496,1497,1498,1499,1500,1501,1502,1503,1504,1505,1506,1507,1508,1509,1510,1511,1512,1513,1514,1515,1516,1517,1518,1519,1520,1521,1522,1523,1524,1525,1526,1527,1528,1529,1530,1531,1532,1533,1534,1535,1536,1537,1538,1539,1540,1541,1542,1543,1544,1545,1546,1547,1548,1549,1550,1551,1552,1553,1554,1555,1556,1557,1558,1559,1560,1561,1562,1563,1564,1565,1566,1567,1568,1569,1570,1571,1572,1573,1574,1575,1576,1577,1578,1579,1580,1581,1582,1583,1584,1585,1586,1587,1588,1589,1590,1591,1592,1593,1594,1595,1596,1597,1598,1599,1600,1601,1602,1603,1604,1605,1606,1607,1608,1609,1610,1611,1612,1613,1614,1615,1616,1617,1618,1619,1620,1621,1622,1623,1624,1625,1626,1627,1628,1629,1630,1631,1632,1633,1634,1635,1636,1637,1638,1639,1640,1641,1642,1643,1644,1645,1646,1647,1648,1649,1650,1651,1652,1653,1654,1655,1656,1657,1658,1659,1660,1661,1662,1663,1664,1665,1666,1667,1668,1669,1670,1671,1672,1673,1674,1675,1676,1677,1678,1679,1680,1681,1682,1683,1684,1685,1686,1687,1688,1689,1690,1691,1692,1693,1694,1695,1696,1697,1698,1699,1700,1701,1702,1703,1704,1705,1706,1707,1708,1709,1710,1711,1712,1713,1714,1715,1716,1717,1718,1719,1720,1721,1722,1723,1724,1725,1726,1727,1728,1729,1730,1731,1732,1733,1734,1735,1736,1737,1738,1739,1740,1741,1742,1743,1744,1745,1746,1747,1748,1749,1750,1751,1752,1753,1754,1755,1756,1757,1758,1759,1760,1761,1762,1763,1764,1765,1766,1767,1768,1769,1770,1771,1772,1773,1774,1775,1776,1777,1778,1779,1780,1781,1782,1783,1784,1785,1786,1787,1788,1789,1790,1791,1792,1793,1794,1795,1796,1797,1798,1799,1800,1801,1802,1803,1804,1805,1806,1807,1808,1809,1810,1811,1812,1813,1814,1815,1816,1817,1818,1819,1820,1821,1822,1823,1824,1825,1826,1827,1828,1829,1830,1831,1832,1833,1834,1835,1836,1837,1838,1839,1840,1841,1842,1843,1844,1845,1846,1847,1848,1849,1850,1851,1852,1853,1854,1855,1856,1857,1858,1859,1860,1861,1862,1863,1864,1865,1866,1867,1868,1869,1870,1871,1872,1873,1874,1875,1876,1877,1878,1879,1880,1881,1882,1883,1884,1885,1886,1887,1888,1889,1890,1891,1892,1893,1894,1895,1896,1897,1898,1899,1900,1901,1902,1903,1904,1905,1906,1907,1908,1909,1910,1911,1912,1913,1914,1915,1916,1917,1918,1919,1920,1921,1922,1923,1924,1925,1926,1927,1928,1929,1930,1931,1932,1933,1934,1935,1936,1937,1938,1939,1940,1941,1942,1943,1944,1945,1946,1947,1948,1949,1950,1951,1952,1953,1954,1955,1956,1957,1958,1959,1960,1961,1962,1963,1964,1965,1966,1967,1968,1969,1970,1971,1972,1973,1974,1975,1976,1977,1978,1979,1980,1981,1982,1983,1984,1985,1986,1987,1988,1989,1990,1991,1992,1993,1994,1995,1996,1997,1998,1999,2000,2001,2002,2003,2004,2005,2006,2007,2008,2009,2010,2011,2012,2013,2014,2015,2016,2017,2018,2019,2020,2021,2022,2023,2024,2025,2026,2027,2028,2029,2030,2031,2032,2033,2034,2035,2036,2037,2038,2039,2040,2041,2042,2043,2044,2045,2046,2047,2048,2049,2050,2051,2052,2053,2054,2055,2056,2057,2058,2059,2060,2061,2062,2063,2064,2065,2066,2067,2068,2069,2070,2071,2072,2073,2074,2075,2076,2077,

O conteúdo das consultas DNS é visível?

Sim, inteiramente. O DNS tradicional trafega em texto claro sobre UDP sem qualquer criptografia. No Wireshark é possível ver sem esforço: o domínio consultado, o tipo de registro, os IPs retornados, o RCODE, os TTLs e eventuais CNAMEs. Um intermediário — ISP, administrador de rede ou atacante em posição de man-in-the-middle — consegue extrair o histórico completo de navegação do usuário, injetar respostas falsas e aplicar bloqueios seletivos.

Parte 3 — DNS over TLS

3.1 Comparação de Desempenho

90 consultas por servidor DoT (10 por domínio × 9 domínios). Tabela comparativa com os equivalentes UDP (valores em ms):

Servidor	UDP Mín	UDP Méd	UDP Máx	DoT Mín	DoT Méd	DoT Máx
Google	52,7	149,7	409,6	208,9	302,3	709,8
Cloudflare	14,4	44,9	700,7	100,6	128,4	393,7
Quad9	12,2	66,3	104,5	63,1	76,5	308,9

Servidor DoT	Suces so	Mín (ms)	Méd (ms)	Máx (ms)
Google DoT (dns.google)	90/90	208,9	302,3	709,8
Cloudflare DoT (one.one.one.one)	90/90	100,6	128,4	393,7
Quad9 DoT (dns.quad9.net)	90/90	63,1	76,5	308,9

Questões para Reflexão

1. O DNS tradicional (UDP, porta 53) oferece privacidade ao usuário?

Não. As consultas trafegam em texto claro, sem autenticação nem criptografia. Qualquer nó intermediário pode ler, modificar ou bloquear as mensagens. O experimento demonstrou isso: o gateway doméstico alterou silenciosamente a resposta de `www.google.com` para `216.239.38.120` (SafeSearch forçado) sem notificar o usuário.

2. O bloqueio por DNS é eficaz como mecanismo de censura?

É eficaz apenas contra usuários leigos. Qualquer usuário técnico pode contorná-lo configurando um servidor DNS alternativo (`1.1.1.1`, `8.8.8.8`) ou usando DoT/DoH. O caso de `polymarket.com` ilustra isso: nenhum servidor testado o bloqueou, e mesmo que o ISP bloqueasse, bastaria trocar o servidor DNS configurado no sistema.

3. Qual a diferença entre um DNS público e um DNS de ISP?

DNS públicos (Google, Cloudflare, Quad9) são operados por empresas de tecnologia, com políticas de privacidade publicadas e variantes com filtragem opcional. DNS de ISPs são operados pelo provedor de acesso, sujeitos a bloqueios judiciais locais e podem registrar consultas sem transparência. O servidor em uso atualmente é o gateway doméstico, que repassa ao DNS do ISP local.

4. Se dois servidores retornam IPs diferentes para o mesmo domínio, como determinar qual é legítimo?

Pelo consenso entre múltiplos servidores independentes: se a maioria retorna o mesmo IP, esse é o resultado legítimo. O IP divergente deve ser investigado — pode indicar redirecionamento (SafeSearch, bloqueio) ou DNS spoofing. DNSSEC permite validar criptograficamente a autenticidade da resposta.

5. Qual abordagem apresenta menor latência: DNS/UDP ou DNS/TLS? Qual apresenta maior overhead?

UDP apresenta menor latência em todos os casos. Google UDP médio: 149,7 ms vs. DoT: 302,3 ms (+102%). Cloudflare UDP: 44,9 ms vs. DoT: 128,4 ms (+186%). Quad9 UDP: 66,3 ms vs. DoT: 76,5 ms (+15%). O overhead do DoT vem do handshake TLS (negociação de certificados e cipher suite) e do uso de TCP em vez de UDP, adicionando 1 a 2 RTTs adicionais por sessão.

6. O conteúdo das consultas DNS é visível no tráfego DoT capturado pelo Wireshark?

Não. No Wireshark o tráfego DoT aparece como TLSv1.3 na porta 853 — os dados são completamente opacos. Domínio consultado, IPs retornados e RCODE são invisíveis (Figura 3). O único dado visível a um observador externo é o IP do servidor DNS de destino e o volume de tráfego.

7. O DoT é mais fácil ou mais difícil de bloquear do que o DNS tradicional?

Tecnicamente mais fácil: basta bloquear a porta TCP 853 no firewall. O DNS/UDP opera na porta 53, universalmente aberta e necessária para o funcionamento básico da rede. Porém, bloquear a 853 é uma ação explícita e identificável, enquanto interferir com UDP/53 pode ser feito de forma silenciosa e seletiva por domínio.

8. Qual protocolo oferece maior privacidade ao usuário?

DoT. As consultas são cifradas com TLS 1.3, impedindo que intermediários leiam ou modifiquem o conteúdo. DNS/UDP não oferece qualquer privacidade — as capturas da Parte 2 demonstram isso: domínio, tipo de registro e IP retornado são completamente legíveis sem nenhuma decifração.